Quantification: 543_z13
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	11
interior lacuna count	11
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1862	1736	823.012	px^2
major_axis_length_px	lacuna shape	78.3615	74.0218	22.3719	px
minor_axis_length_px	lacuna shape	30.8137	31.6227	6.9101	px
aspect_ratio	lacuna shape	2.5873	2.6086	0.6752	unitless
eccentricity	lacuna shape	0.9006	0.9236	0.0676	unitless
solidity	lacuna shape	0.9037	0.9231	0.0388	unitless
roots_count	headline	7.4545	8	2.7336	count
ring_length_r30_px	headline	351.182	368	62.7404	px
ring_length_r60_px	reported	971.636	1007	137.555	px
owned_length_px	ownership-dependent	3223.77	3196.42	1466.07	px
edge_count	ownership-dependent	141.818	135	70.4469	count
mean_edge_length_px	ownership-dependent	23.7241	22.5672	3.4393	px
perimeter_px	normalised	207.479	205.037	54.0901	px
ring_area_r30_px2	normalised	8447.82	8219	1466.9	px^2
ring_area_r60_px2	normalised	21999.7	21794	3275.21	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0423647	0.0375127	0.0096717	px^-1
ring_density_r60	normalised	0.0446693	0.0443017	0.00690422	px^-1
roots_per_100px_perimeter	normalised	3.6357	3.5742	1.2776	per 100 px
ring_attached_length_r30_px	network v2	308.454	318	77.867	px
ring_attached_length_r60_px	network v2	771.182	749	237.268	px
ring_length_w_r30_px	network v2	398.431	419.505	69.0766	px
ring_length_w_r60_px	network v2	1099.87	1126.98	155.79	px
sholl_crossings_r10	network v2	8.5455	9	2.0671	count
sholl_crossings_r20	network v2	11.7273	12	2.4121	count
sholl_crossings_r30	network v2	13.3636	13	2.7303	count
ring_width_mean_r30_px	width	7.3107	7.4604	0.4988	px

Per field
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measure	value	units
analysed_area_px2	1.02809e+06	px^2
total_skeleton_length_px	43175	px
canalicular_length_density_per_px	0.0419952	px^-1
junction_count	992	count
junction_density_per_px2	0.00096489	px^-2
lacuna_count	11	count
lacunae_per_px2	1.07e-05	px^-2
skeleton_component_count	213	count

mean_component_length_px	202.7	px
median_component_length_px	23	px
field_length_density_w_per_px	0.0476972	px^-1
field_density_without_flagged_per_px	0.0419952	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	256	px
unattached_component_count	13	count
canalicular_length_density_attached_per_px	0.0419952	px^-1
field_length_density_with_unattached_per_px	0.0422442	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7697	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10	px
width_pixels_used	43175	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2019	79.1535	33.2822	2.3783	0.9073	0.9417	0.4378	97.9678	270.286	False
2	839	61.6665	17.863	3.4522	0.9571	0.906	0.068	131.349	828.439	False
3	1736	73.3356	33.3815	2.1969	0.8904	0.8374	0.2312	218.82	919.248	False
4	1138	46.8317	31.6227	1.481	0.7376	0.9252	0.396	263.104	340.498	False
5	2267	87.9083	33.2852	2.6411	0.9255	0.9473	0.0042	459.621	481.093	False
6	860	45.4745	26.4762	1.7176	0.813	0.854	0.07	475.221	748.086	False
7	2150	103.918	27.5311	3.7745	0.9643	0.9231	0.2711	546.673	873.734	False
8	3069	109.359	37.084	2.9489	0.9407	0.9099	0.049	642.67	188.33	False
9	1616	74.0218	28.3763	2.6086	0.9236	0.9245	-0.0477	731.536	630.238	False
10	1438	72.5092	25.5367	2.8394	0.9359	0.9236	0.0953	793.137	478.379	False
11	3350	107.799	44.5116	2.4218	0.9108	0.8475	-0.088	854.917	288.104	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	4	288	833	3573.51	167	21.3983	7.513
2	10	407	1013	4964.68	216	22.9846	7.046
3	9	379	858	2549.42	117	21.7899	6.9607
4	3	250	870	1309.37	56	23.3816	7.7258
5	8	364	1094	5461.26	242	22.5672	7.9626
6	6	410	1007	2678.71	132	20.2932	7.4604
7	9	368	1083	3196.42	144	22.1974	6.5711
8	9	370	1037	3504.59	135	25.96	7.5958
9	5	289	878	2268.98	80	28.3623	7.0944
10	7	290	783	1001.85	32	31.3078	6.5541
11	12	448	1232	4952.72	239	20.7227	7.9333

Normalised per lacuna

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lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	205.037	8499	22530	1	1	0.0338863	0.0369729	1.9509
2	151.297	6982	17707	1	1	0.0582927	0.057209	6.6095
3	205.037	8219	20045	1	1	0.0461127	0.0428037	4.3895
4	139.681	6734	18999	1	1	0.037125	0.0457919	2.1477
5	223.823	8900	23453	1	1	0.0408989	0.0466465	3.5742
6	141.338	6516	17817	1	1	0.062922	0.0565191	4.2451
7	254.936	9810	24446	1	1	0.0375127	0.0443017	3.5303
8	278.593	10387	26403	1	1	0.0356214	0.0392758	3.2305
9	195.723	8083	21794	1	1	0.0357541	0.0402863	2.5546
10	184.995	7835	21305	1	1	0.0370134	0.0367519	3.7839
11	301.806	10961	27498	1	1	0.0408722	0.0448033	3.9761

Network v2 per lacuna

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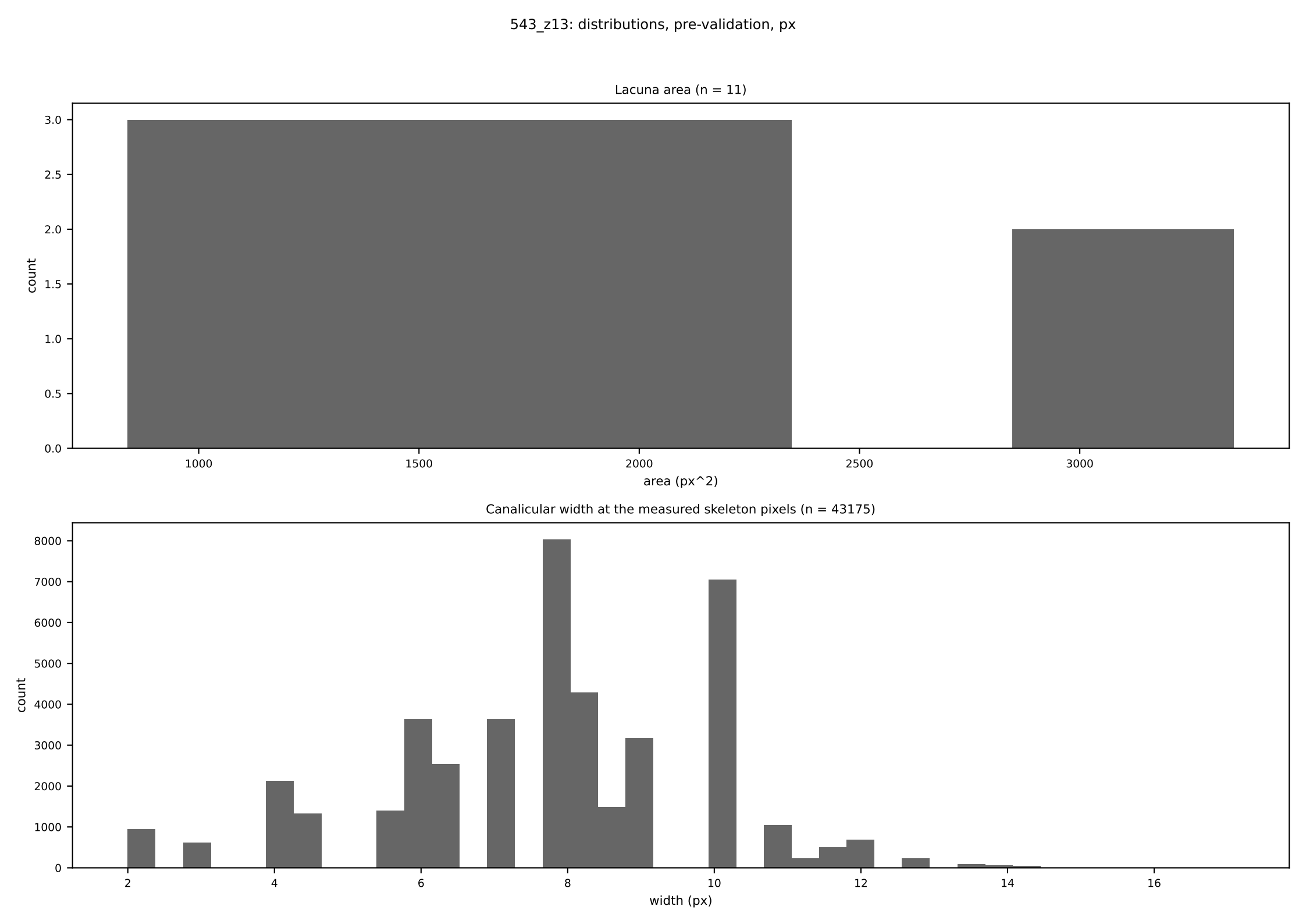
lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	271	658	343.09	961.649	9	11	9
2	397	898	450.593	1127.35	9	13	15
3	296	678	434.505	973.779	9	14	10
4	157	470	289.764	985.808	4	8	13
5	318	903	406.078	1215.95	9	12	17
6	377	846	454.25	1126.98	7	13	14
7	336	749	419.505	1228.6	10	12	13
8	364	1017	425.505	1199.47	10	12	17
9	253	704	326.451	998.122	8	8	12
10	221	364	319.966	876.24	7	10	11
11	403	1196	513.032	1404.61	12	16	16


543_z13: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 43175)
count
width (px)